

How Far Do Subsidized Families Travel to Child Care?

Excess Commuting in Alabama’s Child Care Subsidy System

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ABSTRACT. Families who receive child care subsidies often travel past providers close to home, and the administrative records that document their enrollment say little about why. We apply the excess-commuting framework to Alabama’s subsidy system, comparing the travel that 38,069 subsidized children undertake with the minimum that the locations, licensed capacities, and age restrictions of participating providers would permit. Every observed home-to-provider trip is routed over the road network, and the minimum is obtained by solving capacity-constrained transportation problems. Observed one-way travel averages 13.4 minutes against a constrained minimum of 5.4 minutes, so about 60% of observed travel time is in excess of what the spatial configuration of supply would require. The estimate changes little when providers along commuting corridors are treated as nearly costless detours, when origins are shifted from census tracts to block groups, and when demand is repeatedly resampled. Contrary to what a rural-deficit account would predict, the excess among participating families is largest for infants and in metropolitan Birmingham rather than in the rural Black Belt. Revealed use of providers declines exponentially with travel time, with a half-life of about 6.5 minutes, but retains appreciable weight well beyond the 15-minute catchments that accessibility studies commonly impose. We argue that excess commuting should be read as a diagnostic of unrealized proximity rather than as evidence that reassigning children toward the minimum would improve welfare, and we discuss what the age and geographic patterns imply for subsidy administration and for catchment conventions in accessibility measurement.

Keywords: child care; excess commuting; spatial mismatch; accessibility; travel time; subsidies

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1. Introduction

Child care subsidies reduce the price of care for low- and moderate-income working families, but they do not by themselves determine which providers a family can reach. A provider close to home may not serve the child’s age, may have no open slot, or may fall short of the family’s expectations for quality, hours, or trust, whereas a distant provider may be attractive precisely because it lies along a parent’s journey to work. Geographical access to child care has long been recognized as consequential for parental employment (van Ham & Mulder, 2005), and access measurement in early childhood research has accordingly moved from counts of supply within administrative areas toward family-centered representations of origins, destinations, and travel costs (Blumenberg et al., 2023; Davis et al., 2019; Fransen et al., 2015; Paschall et al., 2021). Travel behavior research has moved in a complementary direction, treating the journey to child care as a distinct and consequential trip purpose rather than an appendage of the commute (Blumenberg et al., 2024).

These two developments leave open a question that neither accessibility indices nor travel-diary descriptions answer directly: how much of the travel that subsidized families actually undertake exceeds the minimum that the existing configuration of homes and provider capacity could support? The excess-commuting framework was developed to answer this question for the journey to work. Observed commutes were first compared with the predictions of a monocentric city model (Hamilton & Röell, 1982), the theoretical benchmark was then replaced by a linear program that reassigns workers to jobs given the actual geography of residences and workplaces (Small & Song, 1992; White, 1988), and the approach was subsequently generalized into a family of benchmarks and indices (Horner, 2002; Kanaroglou et al., 2015; Ma & Banister, 2006; Murphy & Killen, 2011). The resulting minimum is not a forecast or a proposed assignment but a lower bound under stated constraints, and the difference between observed and minimum travel aggregates everything (preference, information, vacancy, schedule, price, and constraint) that keeps realized travel above that bound. Horner and O’Kelly (2007) argued that the same logic extends naturally to non-work travel, applications to school travel followed (Boussauw et al., 2012, 2014; Cheng et al., 2017), and Hu et al. (2024) recently brought the framework to subsidized child care using enrollment records from Florida’s School Readiness program, finding that 51.3% of observed home-to-provider distance was in excess of the feasible minimum.

Three issues raised by that first child care application motivate the present study. The first concerns measurement. The Florida analysis measured road-network distances between tract centroids (with an intrazonal-radius adjustment) and defined supply by observed enrollment, so that families’ revealed choices are partly built into the constraint; whether the conclusion survives routed travel times, an independently measured capacity constraint, and age-eligibility restrictions is not known. A second issue is the anchor. A home-anchored minimum may overstate excess when care is chosen on the way to work, a standard criticism of home-based benchmarks (Fransen et al., 2015; Guyol et al., 2023; Hu & Li, 2021) that carries particular force for child care, because escorting children is commonly embedded in parents’ work journeys and divided within households (McDonald, 2008; Schwanen, 2007). The third issue concerns behavioral calibration. Accessibility studies of child care conventionally impose hard catchments of 10, 15, or 20 minutes, or steep parametric decay (Davis et al., 2019; Luo & Qi, 2009; Negm & El-Geneidy, 2026), although the choice of impedance has long been known to be consequential for accessibility measurement (Handy & Niemeier, 1997); the travel-time decay that subsidy families themselves reveal, measured against the opportunities actually available to them, has not been estimated.

In this paper we address all three issues in a statewide analysis of Alabama’s child care subsidy system, built from restricted administrative records of 38,069 child–provider dyads linked to the state licensing registry and routed on a frozen OpenStreetMap network. Alabama combines metropolitan corridors with extensive non-metropolitan territory, including the historically underserved Black Belt, and is therefore a setting in which no single proximity rule is likely to fit. We solve capacity- and age-constrained transportation problems at two areal scales, quantify sampling uncertainty by re-solving the full optimization in 1,000 bootstrap replicates, re-solve under a work-corridor cost definition constructed from commuting-flow data, obtain stratified estimates by age group, funding category, and geography, and estimate the travel-time decay of revealed use against age-eligible opportunity. To our knowledge, no excess-commuting study of child care has previously reported sampling uncertainty for an excess index, tested areal-unit sensitivity within a single design (a sensitivity urged at least since Horner and Murray (2002) and Niedzielski et al. (2013)), or calibrated travel-time decay against revealed subsidy use. The fitted decay parameters are exported with the replication materials so that subsequent accessibility work can replace imposed catchments with an empirical calibration.

The plan of the paper is as follows. Section 2 places the study in the excess-commuting and child care access literatures. Section 3 describes the administrative data, the routing infrastructure, and the optimization and inference methods. Section 4 presents the statewide estimate and its robustness, stratified estimates by age, geography, and funding category, and the estimated decay of revealed use. Section 5 discusses interpretation, the comparison with Florida, implications for policy and measurement, and limitations. Section 6 concludes.

2. Excess commuting and the journey to child care

2.1. The excess-commuting framework. Hamilton and Röell (1982) compared observed commutes with the minimum implied by a monocentric city and found observed travel roughly eight times the model’s prediction. White (1988) replaced the theoretical benchmark with a linear program that reassigns workers to jobs, given the actual geography of residences and workplaces, and found much smaller excess; Small and Song (1992) consolidated the debate and established the transportation-problem formulation that has anchored the literature since. Later work extended the framework with additional benchmarks, including maximum and random commutes, commuting potential, and commuting economy (Horner, 2002; Murphy & Killen, 2011), and critical reviews consolidated its interpretation: excess commuting indexes the joint outcome of urban form and behavior, not individual irrationality (Kanaroglou et al., 2015; Ma & Banister, 2006). Disaggregated applications show, moreover, that aggregate indices can conceal considerable heterogeneity across population subgroups (Jing & Hu, 2022), which motivates the stratified estimates reported below.

Two findings from this literature shape the design of our study. One is that excess-commuting metrics are sensitive to the areal units at which origins and destinations are aggregated, because coarser zones absorb intrazonal travel and mechanically depress minimum benchmarks (Horner & Murray, 2002; Hu & Wang, 2015; Niedzielski et al., 2013; Openshaw, 1984). A new application should therefore test scale dependence rather than assume it away. The other is that benchmark definitions embed anchor assumptions. Observed commutes reflect trip chains and multi-anchor days, so a home-anchored minimum can overstate excess for trips that are in fact detours on another journey (Hu & Li, 2021). Both points have been raised as limitations of the child care application (Hu et al., 2024) but have not yet been addressed within one design.

2.2. Extensions to school and child care travel. Horner and O’Kelly (2007) asked whether non-work travel is excessive and anticipated the transfer of commuting benchmarks to other obligatory trip purposes. School travel provided the first applications: excess travel in Flemish home-to-school trips (Boussauw et al., 2012, 2014), kindergarten travel in Changchun (Cheng et al., 2017), equity-oriented analyses of school-travel distances in Sweden (Andersson et al., 2012), and descriptive studies of preschool commuting in metropolitan Stockholm (Fjellborg & Forsberg, 2022). These studies established that obligatory child-serving trips are long relative to feasible minima. However, most compared observed travel with nearest-facility assignment, which ignores capacity and eligibility, or worked at a single areal scale.

The journey to child care differs from the journey to school in ways that make those simplifications more consequential. Capacity constraints bind tightly, since infant slots in particular are scarce relative to demand (Kawabata, 2014), and subsidy systems add eligibility rules, copayments, and administrative geography on top of licensing (Herbst & Tekin, 2012; Sandstrom et al., 2018). Proximity is consistently reported as a leading consideration in provider choice, particularly for low-income families with constrained mobility (Dussaillant, 2016; Preston & Rajé, 2007; Shapiro et al., 2019), yet realized enrollment persistently bypasses nearby options (Sandstrom et al., 2018). Hu et al. (2024) formalized the resulting question with a capacity-constrained transportation problem over Florida School Readiness enrollment. Observed mean distance was 9.24 km against a 4.50 km minimum (51.3% excess); 13% of observed but 41% of optimal assignments stayed within the home tract; and infants showed what the authors called a “dual disadvantage”, with both the longest observed and the longest minimum distances.

2.3. Open issues in the child care application. These issues can be stated more precisely as follows. (a) *Measurement*: distance between zone representations is a coarse proxy for the door-to-door road travel that families experience, even though network-based measurement is standard in child care access mapping (Langford et al., 2019); excess estimates in time on a routed network are missing. (b) *Supply definition*: measuring capacity by enrolled children makes supply partly endogenous to the choices under study, whereas licensing records provide a constraint that exists independently of realized enrollment, albeit with measurement limits of their own (Miller-Bains et al., 2025). (c) *Anchor and scale*: the home-only anchor and single-scale aggregation critiques apply with full force, and to our knowledge no child care application has quantified either or reported sampling uncertainty for any excess

index. (d) *Behavioral calibration*: accessibility measurement for child care continues to impose catchment cutoffs and decay forms, such as the three-zone step of Luo and Qi (2009), the steep polynomial-exponential decay of Davis et al. (2019), or hard 15-minute boundaries, whose behavioral content is untested against revealed subsidy use, although recent work has begun to ask what thresholds travelers themselves consider sufficient (Iacono et al., 2010; Negm & El-Geneidy, 2026). The present study addresses each of them.

3. Data and methods

3.1. Setting and administrative dyads. Alabama’s Child Care Subsidy Program, administered by the Alabama Department of Human Resources (DHR) through nine regional service areas, supports care for eligible low- and moderate-income families engaged in work, education, or training (Alabama Department of Human Resources, 2026). Under a restricted data sharing agreement (identifier on the title page), DHR provided a linked administrative extract of subsidized children and their attended providers. The canonical file contains 41,101 child–provider dyads. We retain 38,069 dyads (92.6%) with complete geocoded home and provider coordinates inside the official 2020 Alabama boundary; 3,031 records lack at least one coordinate and one falls outside the state polygon. Each dyad is an observed one-way home-to-attended-provider pairing, not a verified daily trip or a trip chain. The retained sample comprises 2,032 infants (younger than 1), 10,020 toddlers (1 to under 3), 15,540 preschoolers (3 to under 6), and 10,477 school-age children (6 and older). Funding categories comprise At Risk (33,063), Protective Services (1,503), Foster Care (1,389), JOBS (1,221), Early Head Start (515), and Other (378). These categories are administrative eligibility pathways: At Risk covers low-income working families, JOBS families in welfare-to-work activities, Protective Services and Foster Care child-welfare-involved placements, and Early Head Start partnership slots.

The design is governed throughout by disclosure requirements. Exact home coordinates are used only inside the restricted environment, for routing and point-in-polygon assignment; all reported results are aggregates, cells with fewer than 10 observations are suppressed, and maps display no child locations. The online supplement documents the data pipeline, linkage, and disclosure protocol (Sections A and I).

3.2. Participating supply, capacity, and age eligibility. The statewide licensing registry lists 2,167 facilities, 2,164 of them with routable coordinates. The analysis universe is the 1,237 licensed facilities attended by at least one retained subsidy child

(995 centers, 126 family homes, and 116 group homes), which together report 81,055 licensed day slots (Fig. 1). We treat licensed day capacity as a shared facility-level constraint and impose registry-based age eligibility for infant/toddler, preschool, and school-age care, so that a child can be assigned only to a facility licensed for the child’s age band.

Licensed capacity is a regulatory ceiling rather than a count of vacancies or subsidy-dedicated slots, and authorized capacity can diverge from operating capacity (Miller-Bains et al., 2025). In our linkage, 125 facilities record more subsidy dyads than licensed day slots, which is consistent with snapshot timing differences and churn. We nevertheless prefer licensing to enrollment as the constraint because it is measured independently of the choices being evaluated; defining supply by enrollment, as in Hu et al. (2024), guarantees that the observed assignment is feasible but builds revealed behavior into the benchmark. A reconciled-capacity sensitivity is reported with the results (Section 4.2), and capacity measurement is examined further in Section B.

3.3. Routing, counterfactual geography, and the work-flow proxy. All travel costs are one-way car travel times routed on a frozen OpenStreetMap Alabama extract with the Open Source Routing Machine (OSRM) (Luxen & Vetter, 2011), hosted locally so that no child coordinate leaves the restricted environment. Observed costs route each exact home to its attended provider (38,069 routes, all successful). Counterfactual costs route child-weighted centroids of 1,373 populated 2020 census tracts (the primary scale) and 3,356 block groups (the scale sensitivity) to all 1,237 providers (U.S. Census Bureau, 2026b), retaining pairs within a 165-km prefilter that was verified to contain every routed pair at or below 90 minutes. Routed time and distance correlate at $r = 0.98$. The routing infrastructure, its quality assurance, and an aggregate concordance check against an earlier single-county routing effort are described in Section C.

Household workplaces are not observed. For the work-corridor sensitivity we therefore construct, for each home tract, a worker-weighted centroid of workplace tracts from 2023 LEHD Origin–Destination Employment Statistics (LODES) flows for Alabama residents, including cross-border flows into the four neighboring states (U.S. Census Bureau, 2026a). The 1,373 demand tracts contain 2.0 million LODES workers, 5.3% of whom work out of state. This is an area-level proxy; the median tract sends only 8.5% of its workers to its single largest destination tract, which is why we treat the corridor analysis as a bounding sensitivity rather than as a household-level trip-chain model (Section F).

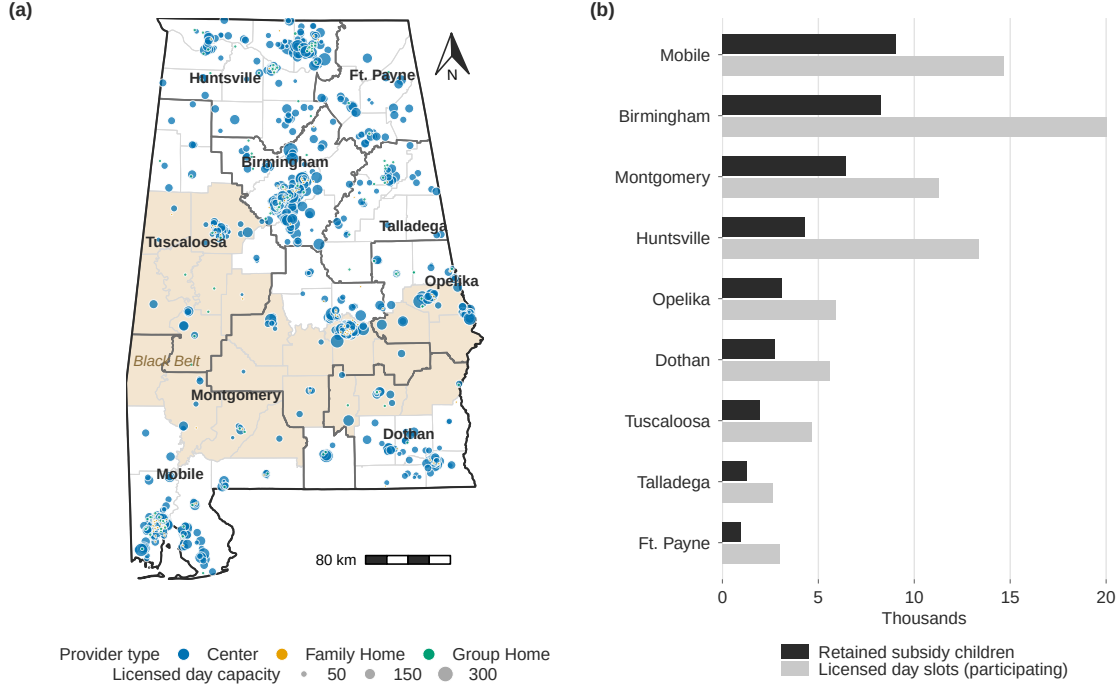


FIGURE 1. Study setting. (a) Subsidy-participating licensed providers (points; public business locations, no child data), the nine DHR service regions (gray boundaries), and the 23-county Black Belt enumeration from Alabama HJR139 (Alabama Legislature, 2025) (shaded). (b) Retained subsidy children and participating licensed day capacity by DHR region. Boundaries: U.S. Census Bureau TIGER/Line 2020. Map lines delineate study areas and do not necessarily depict accepted national boundaries.

3.4. The capacity-constrained excess-commuting model. Let d_{ga} be the number of retained children in origin g and age group a , c_{gj} the routed time from origin g to provider j , K_j provider j 's licensed day capacity, and $E_{aj} \in \{0, 1\}$ age eligibility. The transportation problem is

$$\min_{x_{gaj} \geq 0} \sum_g \sum_a \sum_j c_{gj} x_{gaj} \quad (1)$$

subject to

$$\sum_j x_{gaj} = d_{ga} \quad \forall g, a; \quad \sum_{g,a} x_{gaj} \leq K_j \quad \forall j; \quad x_{gaj} = 0 \text{ if } E_{aj} = 0, \quad (2)$$

where x_{gaj} is the number of children in origin–age cell (g, a) assigned to provider j . The primary tract problem has 5,646,905 decision variables (4,565 origin–age cells $\times$

1,237 providers); because the transportation constraint matrix is totally unimodular, the continuous relaxation returns integer optima. Writing T_{obs} for the observed mean routed time and T_{min} for the optimized mean, absolute excess is $T_{\text{obs}} - T_{\text{min}}$ and the normalized index is

$$T_{\text{ex}} = \frac{T_{\text{obs}} - T_{\text{min}}}{T_{\text{obs}}}, \quad (3)$$

the share of observed mean travel time above the modeled minimum (Kanaroglou et al., 2015; White, 1988).

Three features of the construction bear on interpretation: T_{ex} describes the system rather than individual trips; the optimal allocation preserves aggregate demand and capacity but ignores vacancies, prices, hours, siblings, continuity, and quality; and stratum-specific results (by age, funding category, region, and Black Belt status) are independent marginal transportation problems, each solved against the full fixed capacity, so they are conditional lower bounds that need not aggregate to the joint statewide optimum. Note that decomposing the joint optimum by stratum is not an alternative, because the joint problem admits degenerate optima and stratum shares of joint flows then depend on solver tie-breaking; they are not identified (Section D). As a complementary trip-level diagnostic that imposes no systemwide reallocation, we also compute *bypass*, defined as attended-provider time minus time to the nearest routed, participating, age-eligible provider (Section G).

3.5. Dual-anchor work-corridor sensitivity. A home-only minimum may overstate excess if care is chosen along the work journey (Hu & Li, 2021). For home centroid H_g , provider P_j , and the tract’s worker-weighted work centroid W_g , we define the excess-path detour

$$q_{gj} = \frac{\|H_g - P_j\| + \|P_j - W_g\| - \|H_g - W_g\|}{v}, \quad (4)$$

where $\|\cdot\|$ is Euclidean distance and $v = 50$ km/h converts the detour to minutes; the tolerance grid described below subsumes plausible speed variation because q_{gj} scales inversely with v . At tolerance $\tau \in \{5, 10, 15\}$ minutes, any provider with $q_{gj} \leq \tau$ receives generalized cost $\min(c_{gj}, q_{gj})$, and all other providers keep their home-routed cost. The same rule is applied to each attended provider, so the corridor scenario redefines the observed cost and the cost surface over which the minimum is re-optimized. Because the feasible set only gains low-cost options, T_{min} falls mechanically. Absolute

excess is therefore the comparable quantity across anchors, whereas normalized indices under different cost definitions are different estimands and should not be ranked against each other.

3.6. Scale sensitivity and statistical inference. To test areal sensitivity (Horner & Murray, 2002; Openshaw, 1984), we re-solve Eqs. (1)–(2) with block-group origins (9,215 origin–age cells; 11,398,955 decision variables), holding observed exact-point travel fixed. Intrazonal assignments retain their routed centroid-to-provider times rather than being set to zero, which avoids the mechanical intrazonal advantage that coarser units would otherwise have.

For sampling uncertainty we draw 1,000 bootstrap resamples of the 38,069 dyads. Every replicate rebuilds origin–age demand, recomputes T_{obs} , and re-solves the full transportation problem, together with all 21 stratum problems, against fixed providers, capacities, eligibility, and routed costs. We report percentile 95% intervals. The bootstrap solver reproduces the reference optimum to within 8×10^{-15} minutes (Sections D and E). These intervals quantify demand-resampling variation conditional on measured supply and network; they do not capture capacity, geocoding, or routing measurement error.

3.7. Opportunity-adjusted travel-time decay. Raw travel-time histograms confound willingness to travel with the geography of available supply. We therefore compute, in five-minute bins b of routed time, the utilization rate $u_b = n_b/e_b$, where n_b is the number of observed dyads in bin b and e_b is the demand-weighted count of age-eligible participating-provider opportunities at that time separation. We fit exponential, power, and Gaussian decay forms to the binned counts by Poisson regression with $\log e_b$ as offset, compare forms by the Akaike information criterion (AIC) and log-rate root-mean-square error (RMSE), and scale standard errors by Pearson overdispersion. Fitting uses $0 \leq t < 90$ minutes, where candidate completeness is guaranteed by the routing prefilter; 169 longer observed trips (0.44%) remain in descriptive statistics. For the winning form, relative weight is $w(t) = e^{-\lambda t}$ with half-life $\ln 2/\lambda$. We compare the fitted curve with three conventions from the child care accessibility literature: a Luo–Qi-style three-zone step (1.00 to 5 minutes, 0.68 to 10, 0.22 to 15, zero beyond), which compresses the original primary-care zone boundaries of Luo and Qi (2009) to the shorter catchments conventional in child care accessibility work; a hard 15-minute cutoff; and the $\exp(-t^4/1000)$ decay of Davis et al. (2019) capped at 20 minutes (Section H).

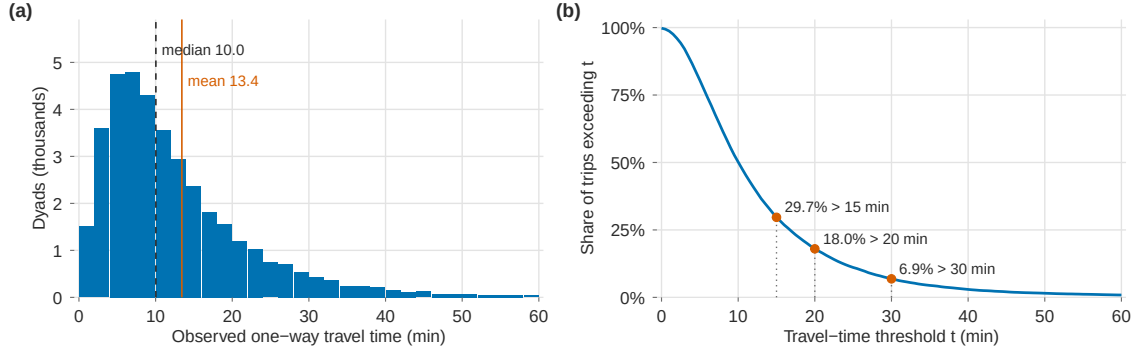


FIGURE 2. Observed one-way home-to-provider travel time ($N = 38,069$). (a) Histogram in two-minute public bins; bins with fewer than 10 dyads are suppressed and the 0.9% of trips beyond 60 minutes are not displayed. (b) Share of trips exceeding each travel-time threshold, with the 15-, 20-, and 30-minute conventions marked.

TABLE 1. Analytic universe, participating supply, and observed travel.

Block	Indicator	Value
Sample	Canonical linked dyads	41,101
	Retained dyads (complete coordinates, in state)	38,069
	Excluded dyads	3,032
Supply	Participating licensed providers	1,237
	Centers / family homes / group homes	995 / 126 / 116
	Licensed day capacity (slots)	81,055
Observed travel	Mean (min)	13.43
	Median (min)	10.04
	95th percentile (min)	33.47
	Share exceeding 15 min (%)	29.7
	Share within home tract (%)	12.2

Note: Dyads are administrative home-to-attended-provider pairings with complete geocoded coordinates inside the 2020 Alabama boundary. Capacity is licensed day capacity of subsidy-participating facilities, not vacancy.

4. Results

4.1. Observed travel. Observed one-way routed travel has mean 13.43 minutes, median 10.04, 90th percentile 26.20, and 95th percentile 33.47, with a long right tail: 29.7% of trips exceed 15 minutes, 18.0% exceed 20, 6.9% exceed 30, and 0.9% exceed 60 (Fig. 2; Table 1). Only 12.2% of dyads attend a provider in their home census tract. Mean routed distance is 11.6 km, which is longer than Florida’s observed mean of 9.24 km (Hu et al., 2024), though the comparison is only indicative given the different metrics and periods.

TABLE 2. Statewide excess commuting and robustness. T_{obs} , T_{min} , and excess in minutes. Corridor rows redefine both the observed and the minimum cost (Eq. 4); normalized indices under different cost definitions are not comparable as rankings.

Specification	T_{obs}	T_{min}	Excess	T_{ex}
Home anchor, tract origins (primary)	13.43	5.40	8.03	59.80%
95% bootstrap CI				59.18–60.14%
Home anchor, block-group origins	13.43	5.47	7.96	59.26%
Home anchor, reconciled capacity	13.43	5.35	8.07	60.14%
Work corridor, $\tau = 5$ min	9.71	1.55	8.16	84.01%
Work corridor, $\tau = 10$ min	8.98	1.34	7.63	85.05%
Work corridor, $\tau = 15$ min	8.67	1.28	7.39	85.21%

Note: 1,000 percentile-bootstrap replicates resample dyads and re-solve the full optimization with supply, capacity, age eligibility, and routed costs fixed. Reconciled capacity is the facility-level maximum of licensed day capacity and observed subsidy enrollment (82,363 slots). Corridor scenarios price any provider whose Euclidean excess-path detour from the home–work axis is at most τ minutes (at 50 km/h) at the smaller of its home-routed and detour cost.

4.2. Statewide excess commuting and robustness. The capacity- and age-constrained minimum is 5.40 minutes. Excess commuting is therefore 8.03 minutes, or $T_{\text{ex}} = 59.80\%$ of observed mean travel time (95% confidence interval [CI] 59.18–60.14%; Table 2). The optimal allocation would place 49.1% of children within their home tract, four times the observed 12.2%. The interval is narrow because it conditions on fixed supply, eligibility, and routed costs; it reflects sampling variation in demand rather than measurement uncertainty in the network or capacity.

A trip-level diagnostic that imposes no reallocation gives qualitatively the same answer. Families attend providers averaging 8.63 minutes beyond their nearest routed, participating, age-eligible option, and 55.0% bypass that nearest option by more than five minutes. Only 13.1% of dyads travel to a provider with a higher observed quality-star tier than their nearest eligible alternative, so measured quality differences alone cannot account for most bypass, though unobserved vacancies, hours, prices, reputation, and sibling or workplace logistics remain candidate explanations (Section G).

4.2.1. Sensitivity to the work-corridor anchor. Under the prespecified $\tau = 10$ corridor, the redefined observed cost falls to 8.98 minutes, since 70.1% of attended providers lie within ten detour minutes of the tract-level home–work axis. However, the re-optimized minimum falls faster, to 1.34 minutes. Absolute excess declines only from 8.03 to 7.63 minutes, and across $\tau \in \{5, 10, 15\}$ it stays within 7.39–8.16 minutes (Table 2; Fig. 3a–b). The normalized index rises to about 85% under the corridor

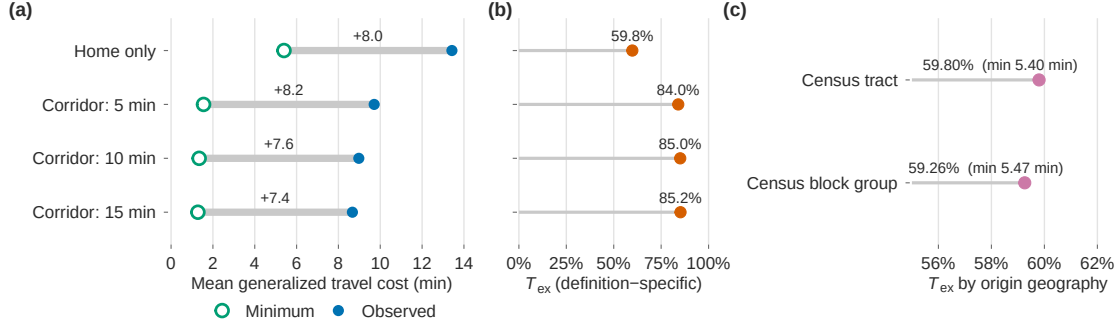


FIGURE 3. Robustness of the excess estimate. (a) Observed and minimum mean generalized cost under the home-only anchor and work-corridor tolerances of 5, 10, and 15 minutes; segment labels give absolute excess in minutes. (b) The normalized index under each cost definition; corridor values are not rank-comparable with the home-only index because the cost definition changes. (c) Tract versus block-group origins.

definition, but this is a denominator artifact of the changed cost definition rather than greater burden, which is why we compare anchors in absolute minutes. Thus even a generous tract-level allowance for work-adjacent care leaves the excess largely intact.

4.2.2. Scale, capacity, and sampling robustness. Re-solving at block-group origins raises the minimum slightly (to 5.47 minutes) and lowers the index to 59.26%, a change of 0.54 percentage points (Fig. 3c). The optimal within-block-group share is 26.2%, and 45.3% of the block-group solution still lands within the home tract, against the tract solution’s 49.1%. The statewide estimate is therefore not an artifact of tract aggregation, which addresses the scale critique of Horner and Murray (2002) within the design, and the bootstrap places tight bounds on sampling variation. Capacity measurement behaves similarly. Re-solving the primary problem under reconciled capacity, which raises the 125 binding facilities to their observed enrollment (Section 3.2), lowers the minimum to 5.35 minutes and moves the index to 60.14% (Table 2). The relaxation that we can measure therefore moves the estimate very little. The direction that we cannot measure (operating capacity below the licensed ceiling) would raise the floor and lower the index, and capacity accordingly remains a stated limitation rather than a resolved one.

4.3. Stratified estimates.

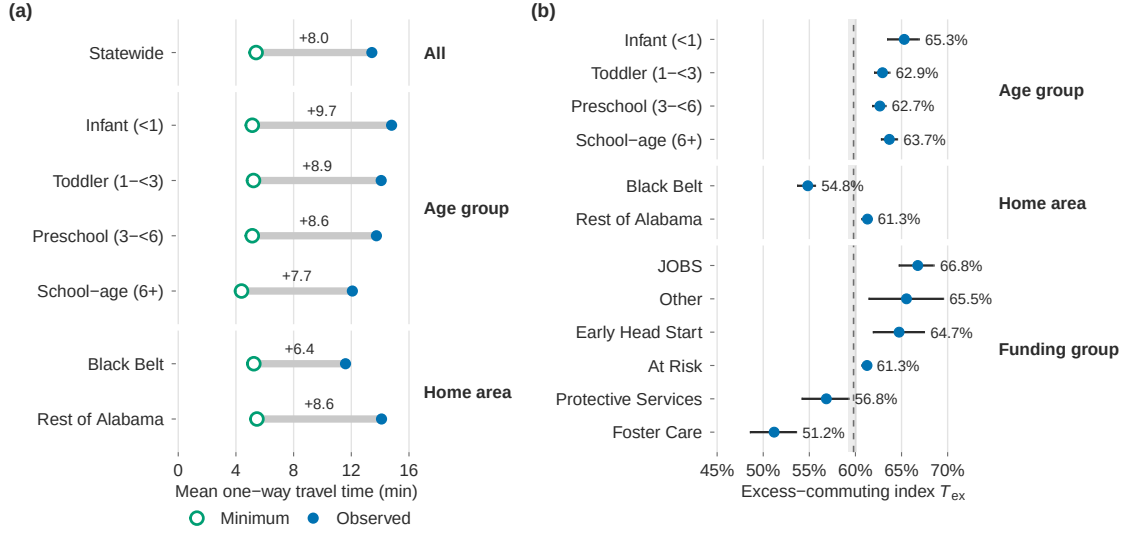


FIGURE 4. Excess commuting by stratum. (a) Observed versus capacity-constrained minimum mean travel time; segment labels give absolute excess in minutes. (b) Normalized index T_{ex} with 95% percentile bootstrap intervals from 1,000 replicates; the dashed line and gray band mark the statewide estimate and its interval. All strata are independent marginal transportation problems.

4.3.1. *Age*. Infants show the longest observed mean (14.79 minutes), the largest absolute excess (9.66 minutes), and the highest stratum index, $T_{ex} = 65.30\%$ (95% CI 63.44–66.88%), against 62.65–63.68% for the other three age groups (Table 3; Fig. 4). Replicate-level paired contrasts substantiate the ordering: the infant index exceeds the toddler index by 2.4 percentage points (95% CI 0.4–4.3) and the preschool index by 2.6 points (0.8–4.4), whereas the contrast with school-age children (1.6 points; -0.4 to 3.4) is not decisive (Section E). The infant minimum (5.13 minutes) is not, however, the highest floor, since toddlers’ minimum is marginally higher (5.22 minutes). Alabama infants therefore carry the largest realized burden relative to what supply geography would permit. This is a different structure from Florida, where infants had both the longest observed and the longest minimum distances and consequently the lowest normalized index of any age group (Hu et al., 2024); we return to the contrast in Section 5.2.

4.3.2. *Geography*. A natural expectation, on supply-scarcity grounds, is that the rural Black Belt would show the greatest realized excess. However, Black Belt residents’ observed mean is shorter (11.60 versus 14.10 minutes), their long tail thinner (20.7%

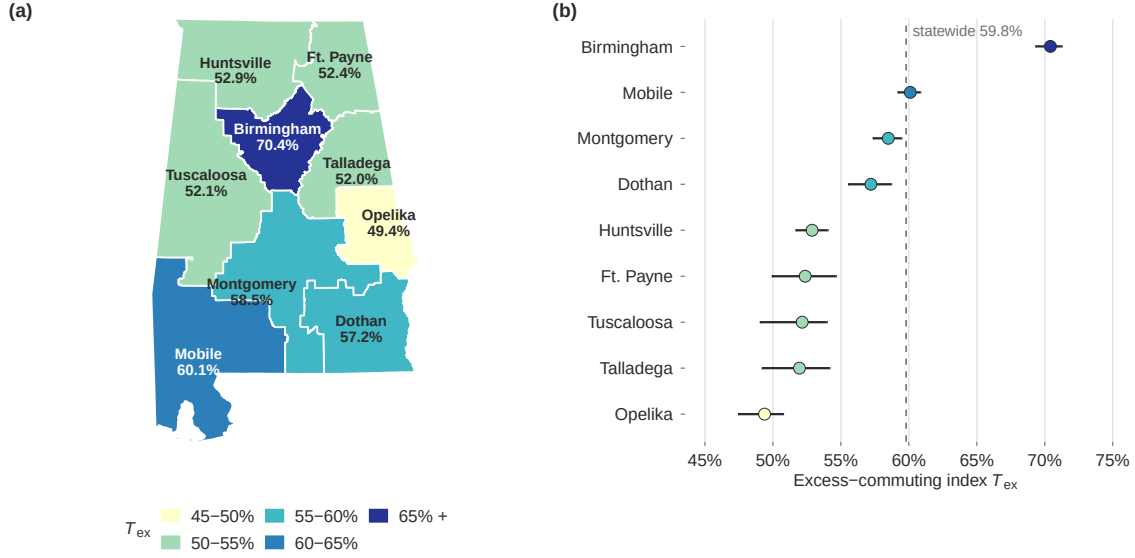


FIGURE 5. Regional geography of excess commuting. (a) Home-region excess index (binned choropleth; region labels give point estimates; no child locations shown). (b) Region estimates with 95% percentile bootstrap intervals; the dashed line marks the statewide index. Boundaries: U.S. Census Bureau TIGER/Line 2020. Map lines delineate study areas and do not necessarily depict accepted national boundaries.

versus 33.0% of trips over 15 minutes), and their excess share lower (54.83%, CI 53.68–55.62%, versus 61.30%, CI 60.65–61.74%) than in the rest of the state. Across the nine DHR service regions, T_{ex} ranges from 49.38% in Opelika to 70.41% in Birmingham, whose interval (69.31–71.25%) is separated from every other region’s (Fig. 5). The separation has a simple structure: among the nine regions, Birmingham pairs the longest observed mean (15.69 minutes) with the lowest minimum floor (4.64 minutes; Section G), so its families travel farthest over the geography that could support the shortest trips. Realized excess in this system is at least as much a metropolitan phenomenon as a rural one.

However, two qualifications apply. The estimates describe realized trips of enrolled participants; families deterred from participating altogether, plausibly more common where supply is thinnest, do not appear, so low realized excess in the Black Belt cannot be read as adequate rural access. Furthermore, the region estimates are independent marginal problems rather than a simultaneous nine-region allocation.

4.3.3. *Funding category.* Funding-group indices span 51.18% (Foster Care, CI 48.55–53.56%) to 66.76% (JOBS, CI 64.70–68.48%), with wider intervals for the small groups

(Table 3). Foster Care’s low index reflects a high floor ($T_{\min} = 7.29$ minutes) rather than short observed travel, which is consistent with placement constraints that are not proximity-driven. These contrasts bundle eligibility, geography, work requirements, and selection; they are descriptive rather than effects of program rules.

4.4. Revealed travel-time decay. The exponential form is clearly preferred for the pooled sample and within every age group (the pooled ΔAIC to the power form exceeds 3,800; Section H). The pooled decay parameter is $\lambda = 0.1062$ per minute (overdispersion-scaled 95% CI 0.0943–0.1181), which corresponds to a half-life of 6.53 minutes (CI 5.87–7.35) and a 5%-weight point at 28.2 minutes (Fig. 6a). Age-specific half-lives decline monotonically with age: 7.71 minutes for infants, 6.99 for toddlers, 6.66 for preschoolers, and 5.60 for school-age children. Relative use of farther opportunities thus decays slowest for the group with the largest excess burden, mirroring the age gradient in Florida (Hu et al., 2024).

The fitted curve retains weight where imposed conventions allow none: 0.35 at 10 minutes, 0.20 at 15, 0.12 at 20, and 0.04 at 30 (Fig. 6b). A hard 15-minute catchment discards the 29.7% of observed trips beyond it; the Davis et al. (2019) decay is far steeper than revealed use (weight 4.5×10^{-5} at 10 minutes); and the Luo–Qi-style step approximates the empirical weight at 15 minutes but reallocates weight toward the first ten minutes and truncates the tail entirely. These are revealed-use calibrations under the current opportunity structure, not statements that families should be expected to travel 28 minutes. They do imply, however, that hard sub-20-minute catchments systematically understate the spatial extent of the market that subsidy families actually use. The fitted parameters are exported with the replication materials for reuse as accessibility weights, with one caveat attached: they are estimated on enrolled participants, and families deterred by distance never appear, so the curve plausibly understates the distance sensitivity of the full eligible population.

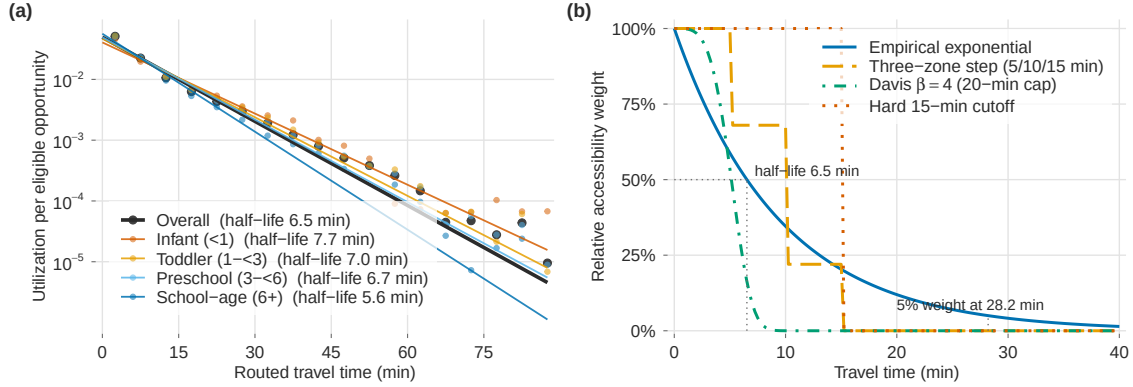


FIGURE 6. Opportunity-adjusted travel-time decay of revealed subsidy use. (a) Utilization per demand-weighted age-eligible opportunity in five-minute bins (points; log scale; bins with zero observed trips omitted from display) with fitted exponential curves; the legend gives fitted half-lives. (b) Fitted relative weight versus three imposed conventions from the child care accessibility literature. Fits use trips below 90 minutes with Poisson regression on an opportunity offset; the sparse bins beyond about 70 minutes carry little exposure weight, which is why they sit above the fitted lines.

TABLE 3. Stratified excess commuting: independent marginal transportation problems with 95% percentile bootstrap intervals.

Dimension	Stratum	N	T_{obs}	T_{min}	Excess	T_{ex} (95% CI)
All	Statewide	38,069	13.43	5.40	8.03	59.80 (59.18–60.14)
Age	Infant (<1)	2,032	14.79	5.13	9.66	65.30 (63.44–66.88)
	Toddler (1–<3)	10,020	14.07	5.22	8.86	62.93 (62.02–63.69)
	Preschool (3–<6)	15,540	13.74	5.13	8.61	62.65 (61.81–63.27)
	School-age (6+)	10,477	12.07	4.39	7.69	63.68 (62.76–64.52)
Area	Black Belt (23 counties)	10,225	11.60	5.24	6.36	54.83 (53.68–55.62)
	Rest of Alabama	27,844	14.10	5.45	8.64	61.30 (60.65–61.74)
Funding	JOBS	1,221	10.40	3.46	6.94	66.76 (64.70–68.48)
	Other	378	15.87	5.47	10.40	65.54 (61.42–69.51)
	Early Head Start	515	15.43	5.44	9.99	64.74 (61.89–67.44)
	At Risk	33,063	13.35	5.17	8.18	61.26 (60.66–61.63)
	Protective Services	1,503	14.87	6.42	8.45	56.84 (54.17–59.25)
	Foster Care	1,389	14.93	7.29	7.64	51.18 (48.55–53.56)

Note: T_{obs} , T_{min} , and excess in minutes; T_{ex} in percent. Each stratum is an independent capacity-constrained transportation problem against the full fixed supply, so strata are conditional lower bounds that need not aggregate to the statewide optimum. Region-level estimates appear in

Fig. 5; full tables in Section G.

5. Discussion

5.1. Interpretation. Alabama subsidy families travel about eight minutes, or roughly 60% of observed mean travel time, beyond a minimum that respects where participating providers are located, how many children they are licensed to serve, and which ages they serve. The estimate moves by less than one percentage point between tract and block-group origins and by 0.4 minutes under a generous work-corridor cost definition, and its bootstrap interval spans less than one percentage point. In the vocabulary of the excess-commuting literature, the realized child care assignment sits far from the system optimum, and that distance is not an artifact of scale, anchor, or sampling.

What the gap means is a separate question, and the classic critique of wasteful-commuting interpretations applies with additional force here (Ma & Banister, 2006; White, 1988). The optimization ignores vacancies, prices and copayments, operating hours, quality beyond the observed star tier, trust, language, siblings, and continuity of care, all of which families weigh deliberately (Guyol et al., 2023; Shapiro et al., 2019). The bypass evidence itself shows that measured quality tiers cannot account for most nearby-provider bypass. This observation works in both directions: families are evidently not simply seeking out rated quality, but neither can the data show that their longer trips are unconsidered. We therefore read the eight-minute gap as *unrealized proximity under measured capacity*, a diagnostic of how much closer the system could in principle place children to care. It is neither a list of children who should move nor evidence that the minimum-travel assignment would raise welfare. The corridor analysis reinforces this diagnostic reading, since the aggregate gap barely moves even when any provider near the home–work axis is priced as nearly free.

5.2. Comparison with the Florida application. The two statewide subsidy systems agree structurally. Excess is large in both (59.8% in time here; 51.3% in distance in Florida), observed within-tract shares are near 12–13% in both, optimal within-tract assignment is several-fold higher (49% versus 41%), and infants show the slowest travel decay in both (Hu et al., 2024). The magnitudes are not comparable as a state ranking, because the designs differ in metric (routed minutes versus network kilometers), in supply definition (licensed capacity versus enrolled children), and in period.

The age structure, however, differs in a way that matters for policy targeting. Florida’s infants faced a dual disadvantage, with the longest observed and the longest

minimum distances, so their normalized index was the lowest of any age group. Alabama’s infants have the longest observed travel but an unexceptional minimum floor; their disadvantage is concentrated in the realized gap (65.3%) rather than in the supply floor. Some of this difference is measurement. Our licensed-capacity constraint with broad age-eligibility flags is more permissive about where infants could go than Florida’s enrollment-based supply, which embeds the scarcity of operating infant slots in the benchmark itself (Kawabata, 2014; Miller-Bains et al., 2025). Read together, the two studies suggest that infant burden has both a floor component (where infant slots exist at all) and a realized component (which feasible slots families can actually use), and that which component dominates depends on how supply is measured. Comparisons across systems should state that dependence rather than carry a single “infant disadvantage” conclusion from one system to another.

5.3. Implications for policy and measurement. For subsidy administration, the results carry three implications. First, infant care warrants attention on both margins. The realized gap (65.3% excess, 9.7 excess minutes, and the slowest decay) points to feasibility barriers among nominally eligible nearby slots, such as vacancies, schedules, and navigation, and not only to slot creation. Second, the geography of intervention should follow the estimates rather than a rural presumption. Birmingham’s index is separated from every other region’s, whereas the Black Belt shows shorter realized travel and lower excess among participants. High metropolitan excess is consistent with dense option sets being bypassed for unobserved reasons (congestion-adjusted feasibility, hours, cross-flows with work geography), and administrative follow-up there has more nearby feasible options to mobilize than a thin-supply reading would suggest. At the same time, the Black Belt result should prompt an analysis of non-participation, which realized-trip data cannot address. Third, reducing travel is not in itself a welfare improvement: policies should widen feasible nearby choice, through slots, hours, information, and transportation, rather than steer families toward the minimum-travel assignment.

For measurement practice, the decay results argue against hard sub-20-minute catchments in child care accessibility work. Nearly 30% of observed subsidized trips exceed 15 minutes, and revealed use retains 12% relative weight at 20 minutes. Where a single parametric weight is needed, the fitted exponential (half-life 6.5 minutes) provides an empirically grounded default, and reporting results under several catchments alongside an empirical weight, in the spirit of threshold-sensitivity work in accessibility research (Iacono et al., 2010; Negm & El-Geneidy, 2026), is preferable to any single

boundary (Handy & Niemeier, 1997). More broadly, excess commuting complements rather than replaces multidimensional access frameworks (Davis et al., 2019; Paschall et al., 2021): it prices one dimension, realized proximity, under explicit constraints, and leaves affordability, quality, and availability to be measured on their own terms.

5.4. Limitations. Several limitations should be borne in mind. (a) The universe is realized subsidy dyads; eligible nonparticipants and unmet demand are absent, which matters most where supply is thinnest. (b) Licensed day capacity is neither vacancy nor subsidy-dedicated capacity, and 125 facilities record more subsidy children than licensed day slots, consistent with snapshot timing differences; age-eligibility flags do not identify age-dedicated slots. (c) Competition from nonsubsidized children is not modeled. (d) The work anchor is a tract-level LODES proxy from 2023, not household workplaces or observed trip chains. (e) The design is cross-sectional and descriptive; no stratum contrast is causal. (f) Bootstrap intervals condition on fixed supply and costs and exclude household clustering and measurement error in capacity, geocoding, and routing. (g) Observed trips use exact coordinates while counterfactual origins use centroids, a privacy-imposed asymmetry that the block-group sensitivity only partially probes. (h) A fixed car profile omits congestion, transit, walking, and ridesharing, a meaningful restriction for low-income households (Preston & Rajé, 2007). (i) The quality tier is incomplete and one-dimensional. Appendices A–I document each point and the associated diagnostics.

6. Conclusions

We have applied the excess-commuting framework to a statewide child care subsidy system, using routed travel times, an independently measured capacity constraint, and age eligibility, together with anchor, scale, and capacity sensitivities and bootstrap inference. Across 38,069 routed subsidy trips in Alabama, observed one-way travel averages 13.43 minutes against a 5.40-minute constrained minimum, so 59.80% of observed travel time is in excess, and the estimate is robust to a work-corridor anchor, to origin scale, and to dyad resampling. Among enrolled families the burden is largest for infants, driven by the realized gap rather than by a uniquely poor supply floor, in contrast with Florida, and it is concentrated in Birmingham rather than in the rural Black Belt. Revealed use decays exponentially with a half-life of about 6.5 minutes yet keeps meaningful weight far beyond the 15-minute catchments that accessibility studies routinely impose.

The analysis shows that the framework transfers to subsidized child care with its modern apparatus intact and returns quantities that accessibility and planning work can reuse: an empirical decay curve and stratum-specific lower bounds. It also shows that unrealized proximity is a large, precisely estimated, and unevenly distributed feature of a statewide subsidy system. The minimum remains a benchmark for diagnosis rather than an assignment to enforce, and interpretation should be confined accordingly. Extensions to nonparticipating eligible families, vacancy-aware capacity, and multimodal costs are the natural next steps.

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Software and reproducibility. The analysis is a fully scripted R pipeline: home-to-provider and counterfactual travel times are routed with the Open Source Routing Machine on a frozen OpenStreetMap network, the capacity-constrained transportation problems are solved with a transportation-simplex solver, sampling uncertainty is quantified by re-solving the full optimization in 1,000 bootstrap replicates, and a reproducibility audit regenerates every reported number from the analytic inputs ([Section I](#)). A public replication package—the analysis code, source manifests, disclosure-safe derived outputs (accessibility and excess statistics, stratum tables, and the fitted travel-time decay parameters), and the publication figures and tables—is available at <https://github.com/joonho112/childcare-excess-commuting-replication>. The child-level administrative records are governed data and are not redistributed in the package.

Data availability. The child-level administrative records used to construct the home-provider dyads and the supply constraint are governed administrative data provided by the Alabama Department of Human Resources under a restricted data sharing agreement and cannot be redistributed by the authors. All other inputs are public: American Community Survey estimates, 2020 Census tract and block-group geography, LEHD Origin–Destination Employment Statistics, and OpenStreetMap road-network data. The replication package supplies the code and disclosure-safe

derived outputs sufficient to reproduce the reported results; aggregate cells with fewer than 10 observations are suppressed throughout.

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Declaration of generative AI use. During the preparation of this work the authors used Claude (Anthropic) as a coding assistant in developing the analysis code. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of this article.

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Online Supplement

How Far Do Subsidized Families Travel to Child Care? Excess Commuting in Alabama’s Child Care Subsidy System

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This supplement accompanies the main text. Section, equation, table, and figure numbers below are prefixed by the appendix letter (A.1, B.1, ...); references to the main text cite it by section, figure, and table number.

Contents

Appendix A. Data sources, linkage, and the analytic universe	27
A.1. Restricted administrative sources	27
A.2. Exclusion cascade	28
A.3. Age bands and funding groups	28
A.4. Geocoding quality assurance	28
A.5. Areal strata: Black Belt and DHR service regions	29
A.6. Demand aggregation for the counterfactual	29
Appendix B. Participating supply and capacity measurement	30
B.1. Supply universe construction	30
B.2. Capacity definitions and the over-capacity diagnostic	30
B.3. Age eligibility	31
Appendix C. Routing infrastructure and validation	31
C.1. Frozen network and engine configuration	31
C.2. Observed routes and candidate matrices	31
C.3. Quality assurance	32
C.4. Aggregate concordance with an earlier single-county routing effort	32
Appendix D. Transportation problems: solver, scale, and estimands	33
D.1. Formulation and solver	33
D.2. Problem sizes and solutions	33
D.3. Why strata are independent marginal problems	33
D.4. Scale sensitivity mechanics	34
Appendix E. Bootstrap inference	35

E.1.	Design	35
E.2.	Fast exact solver and benchmark	35
E.3.	Complete interval table	35
E.4.	Paired stratum contrasts	36
E.5.	History note	37
Appendix F.	Dual-anchor work-corridor sensitivity	37
F.1.	LODES processing and workplace centroids	37
F.2.	Corridor cost construction	38
F.3.	Complete results	38
F.4.	Why anchors are compared in absolute minutes	38
Appendix G.	Complete stratified results and the bypass diagnostic	39
G.1.	Stratified excess commuting with cost decompositions	39
G.2.	Bypass: definition and construction	40
G.3.	Bypass results	41
G.4.	A descriptive bypass model	41
Appendix H.	Opportunity-adjusted travel-time decay	42
H.1.	Exposure construction and likelihood	42
H.2.	Functional-form comparison	43
H.3.	Age-specific parameters	43
H.4.	Comparison with imposed conventions	43
H.5.	Exported parameters	44
Appendix I.	Privacy, disclosure control, and reproducibility	44
I.1.	Governance and the restricted–public boundary	44
I.2.	Computational environment	45
I.3.	Reproducibility and claim verification	45

Appendix A. Data sources, linkage, and the analytic universe

A.1. Restricted administrative sources. The analysis draws on a linked administrative extract provided by the Alabama Department of Human Resources (DHR) under a restricted data sharing agreement (identifier on the title page). The canonical dyad file links each subsidized child to the licensed provider recorded as attended, with pre-geocoded home and provider coordinates supplied by a commercial geocoding vendor upstream of this project. Supporting assets are the statewide licensed-provider registry (2,167 facilities), a curated provider coordinate layer (2,164 facilities with

verified routable coordinates), official 2020 census geography (U.S. Census Bureau, 2026b), 2023 LODES origin–destination employment flows (U.S. Census Bureau, 2026a), and a frozen OpenStreetMap Alabama road-network extract (Appendix C). Every input file, its dimensions, and its cryptographic fingerprint where applicable are registered in the project’s provenance metadata.

The placement extract and the provider-registry snapshot are administrative files maintained on different update cycles and are not perfectly contemporaneous. Each dyad should be read as an observed home-to-attended-provider pairing during the extract’s service window, not as a verified daily trip, a trip frequency, or a household trip chain.

A.2. Exclusion cascade. Table A.1 reports the exclusion cascade from the canonical file to the analytic universe. Of 41,101 canonical dyads, 3,031 lack at least one required coordinate and one additional record geocodes inside the conservative bounding box but outside the official 2020 Alabama state polygon, leaving 38,069 retained dyads (92.6%). No retained record required coordinate repair, and no finite-value or bounds exclusions occurred beyond those shown.

TABLE A.1. Exclusion cascade from canonical linked dyads to the analytic universe

Stage	Remaining	Excluded at stage
Canonical linked dyads	41,101	—
Complete home and provider coordinates	38,070	3,031
Finite coordinates	38,070	0
Within conservative Alabama bounds	38,070	0
Inside official 2020 Alabama boundary	38,069	1
Final analytic universe	38,069	3,032 total

A.3. Age bands and funding groups. Age bands derive from the recorded child age at the extract: Infant (<1 year), Toddler (1 to <3), Preschool (3 to <6), and School-age (6 and older). Source ages span 0.01–16.10 years with median 4.03. Eleven source funding types are recoded to six analysis groups; the small Teen Parent category is folded into At Risk. Table A.2 reports retained counts.

TABLE A.2. Retained dyads by age band and funding group

Stratum	<i>N</i>	Share (%)
Infant (<1)	2,032	5.3
Toddler (1–<3)	10,020	26.3
Preschool (3–<6)	15,540	40.8
School-age (6+)	10,477	27.5
At Risk	33,063	86.9
Protective Services	1,503	3.9
Foster Care	1,389	3.6
JOBS	1,221	3.2
Early Head Start	515	1.4
Other	378	1.0

A.4. Geocoding quality assurance. Coordinates arrive pre-geocoded; we use the vendor’s source census codes only as quality checks, never as join keys. Canonical geography assignment is a local point-in-polygon join of each exact home coordinate against official 2020 cartographic tract and block-group polygons in a projected state coordinate system. Agreement between the vendor’s source tract suffix and the local 2020 spatial join is 99.74%; agreement at the block-group digit is 99.49%. Disagreements concentrate on parcel-scale boundary cases and do not affect any aggregate result at the scales analyzed.

A.5. Areal strata: Black Belt and DHR service regions. The Black Belt stratum uses the inclusive 23-county enumeration that appears in a 2025 legislative resolution (HJR139) (Alabama Legislature, 2025): Barbour, Bullock, Butler, Choctaw, Clarke, Conecuh, Crenshaw, Dallas, Greene, Hale, Lee, Lowndes, Macon, Marengo, Monroe, Montgomery, Perry, Pickens, Pike, Russell, Sumter, Tuscaloosa, and Wilcox. This inclusive list is broader than vernacular definitions of the region and is not synonymous with “rural”; classification in the analysis uses the official 2020 county of the home tract (10,225 Black Belt dyads; 27,844 elsewhere).

The nine DHR service regions (Birmingham, Mobile, Montgomery, Huntsville, Tuscaloosa, Dothan, Opelika, Talladega, Ft. Payne) are county-defined administrative areas. The county-to-region crosswalk is built empirically from provider service-region assignments in the registry; two counties without a retained attended provider (Choctaw, St. Clair) are supplemented manually to their administrative regions (Mobile and Birmingham, respectively). Home-region assignment follows the home tract’s county. Descriptive tabulations that use the extract’s own home-county label instead

of the official home-tract county differ by at most eight borderline dyads for the Black Belt split.

A.6. Demand aggregation for the counterfactual. Counterfactual demand origins are the 1,373 populated 2020 census tracts (of 1,436) and 3,356 populated block groups (of 3,924). Each origin’s representative point is the arithmetic mean of the exact home coordinates of its retained subsidy children (a child-weighted analytic centroid), shared across age and funding strata. Origin-by-age demand tables contain 7,676 tract-level and 12,250 block-group-level strata; both reproduce the 38,069 total exactly, with zero join loss. Strata with fewer than 10 children carry a suppression flag that governs any public release ([Appendix I](#)); the restricted analytic objects retain true counts.

Appendix B. Participating supply and capacity measurement

B.1. Supply universe construction. The statewide licensing registry lists 2,167 facilities with unique identifiers, of which 2,164 have verified routable coordinates in the curated provider layer; the three facilities without coordinates never appear in a retained dyad. The primary analysis universe is the subset of 1,237 licensed facilities attended by at least one retained subsidy child (“observed participation”). Every retained dyad links one-to-one to this universe, and every universe member links one-to-one to the registry and the coordinate layer. [Table B.1](#) summarizes the universe by facility type.

TABLE B.1. Participating supply by facility type

Facility type	Providers	Licensed day slots	Retained dyads
Center	995	78,967	36,594
Family home	126	698	585
Group home	116	1,390	890
Total	1,237	81,055	38,069

B.2. Capacity definitions and the over-capacity diagnostic. The primary constraint is registry *licensed day capacity*: 81,055 slots across the 1,237 participating facilities. Two alternatives are retained as sensitivity inputs: a circular lower bound equal to each facility’s observed subsidy dyad count (total 38,069), and a reconciled capacity equal to the facility-level maximum of the two (total 82,363). We prefer

licensed capacity for the primary estimate because it exists independently of the enrollment choices under study; the reconciled variant is reported as a sensitivity in the main text (Table 2). Re-solving the primary tract problem under reconciled capacity lowers the minimum from 5.397 to 5.351 minutes and raises the index from 59.80% to 60.14%; the measurable relaxation of the constraint therefore moves the estimate by about a third of a percentage point.

A diagnostic motivates the caution: 125 participating facilities (80 centers, 27 family homes, 18 group homes) record more subsidy dyads than licensed day slots. This pattern is consistent with the placement extract and the registry snapshot covering slightly different windows, with enrollment churn within the window, and with capacity updates; it is also consistent with the broader finding that authorized capacity imperfectly tracks operating capacity (Miller-Bains et al., 2025). The transportation problems remain feasible throughout because total capacity comfortably exceeds total demand and every origin–age cell has more eligible candidate capacity than its own demand (Appendix D).

We evaluated state quarterly licensing surveys as a potential facility-level effective-capacity source and rejected them: the published workbooks are state- and county-by-sector aggregates with no facility linking key (provider-level key coverage 0%), so they cannot adjust individual constraints without arbitrary allocation.

B.3. Age eligibility. Registry day-age-range fields define three eligibility flags per facility: infant/toddler, preschool, and school-age. Age-specific slot counts do not exist in the registry, so licensed day capacity is treated as a shared facility total and the optimization excludes ineligible assignments outright ($x_{gaj} = 0$ when $E_{aj} = 0$) rather than splitting capacity across age bands by an arbitrary rule. This is permissive in one direction: a facility flagged for an age band is assumed able to devote any share of its licensed slots to that band. The main text discusses the resulting interpretation limits for age-stratum floors (main text, Section 5.2).

Appendix C. Routing infrastructure and validation

C.1. Frozen network and engine configuration. All travel costs are one-way car travel times computed with the OSRM routing engine (Luxen & Vetter, 2011), version 26.7.3, multi-level Dijkstra pipeline, standard car profile, running against a frozen OpenStreetMap Alabama extract (145,336,067 bytes; MD5 and SHA-256 fingerprints pinned in the provenance register). The server runs on localhost only, so no child coordinate ever leaves the restricted environment. The identical frozen

graph produced every observed route, every counterfactual candidate matrix, and every nearest-provider search, eliminating network-vintage inconsistency across analysis stages.

Production requests use sequential batched table queries in a single process. An earlier parallel implementation was discarded after spot checks revealed cache responses mixing across concurrent tasks (maximum observed error 31.94 minutes); its caches were deleted and all routing was redone sequentially, after which twenty independent one-off re-routes matched batch results exactly (maximum absolute difference 0 minutes).

C.2. Observed routes and candidate matrices. Observed costs route each of the 38,069 exact home coordinates to its attended provider (762 batched requests; zero unreachable pairs; no distance prefilter, so the long tail is preserved; 78 attended pairs exceed 165 km straight-line and are routed anyway). Counterfactual costs route 4,729 origin centroids (1,373 tracts + 3,356 block groups) to all 1,237 providers: 5,849,773 origin-provider pairs in 2,375 batched requests. The canonical candidate set retains the 2,672,355 pairs within a 165-km Euclidean prefilter (110 km/h $\times$ 90 minutes, matching the project’s companion accessibility infrastructure).

C.3. Quality assurance. We applied five checks before accepting the routing outputs. (a) *Prefilter sufficiency*: every pair outside the 165-km prefilter was routed exhaustively; none has a routed time at or below 90 minutes, so the prefilter cannot clip any optimum involving costs up to 90 minutes. (b) *Candidate adequacy*: candidates per origin range 109–857 (median 599), and the minimum origin-level ratio of eligible candidate capacity to own demand is 83.0. (c) *Spot re-routing*: ten observed and ten candidate pairs re-routed one-by-one match batch outputs with zero difference. (d) *Internal consistency*: routed time and routed distance correlate at Pearson $r = 0.980$ (Spearman $\rho = 0.979$). (e) *Distributional plausibility*: [Table C.1](#) summarizes the observed distribution; 169 dyads (0.44%) exceed 90 minutes and 139 (0.37%) exceed 120 minutes, a real but thin tail retained in all descriptive statistics.

TABLE C.1. Observed routed travel: distributional summary ($N = 38,069$)

Quantity	Time (min)	Distance (km)
Mean	13.43	11.59
25th percentile	5.86	3.77
Median	10.04	7.45
75th percentile	16.77	14.05
90th percentile	26.20	24.06
95th percentile	33.47	32.78
99th percentile	57.79	61.04
Maximum	221.78	286.52

C.4. Aggregate concordance with an earlier single-county routing effort.

An earlier restricted case study routed 1,323 subsidy dyads in one county with an independently assembled pipeline. Re-selecting that county from the present statewide universe yields 1,301 dyads (a 22-record difference attributable to linkage and boundary gates). Table C.2 shows the aggregate distributions agree to within a quarter minute at every compared quantile, validating the new statewide routing pipeline at the aggregate level. This is deliberately not a row-level identity check; the two extracts differ slightly in membership.

TABLE C.2. Aggregate concordance between the legacy single-county object and the present statewide pipeline restricted to the same county

Metric	Legacy ($N = 1,323$)	Present ($N = 1,301$)	Difference
Mean (min)	12.17	12.21	0.04
Median (min)	10.00	10.17	0.17
95th percentile (min)	28.48	28.52	0.04
Share above 60 min (%)	0.91	0.92	0.02

Appendix D. Transportation problems: solver, scale, and estimands

D.1. Formulation and solver. The optimization is the classical capacity-constrained transportation problem of the excess-commuting literature (White, 1988), stated in the main text as Eqs. (1)–(2). Ineligible or out-of-candidate origin–provider combinations receive a prohibitive cost (big- M of 10,000 minutes); no optimal solution uses a big- M arc. The primary solver is the transportation-simplex routine of the `lpSolve` interface to the `lp_solve` library, solving the continuous relaxation. Because the transportation constraint matrix is totally unimodular with integer demands and

capacities, optimal vertices are integral; all reported solutions are integer to within 10^{-7} .

D.2. Problem sizes and solutions. Table D.1 records the scale of each solved problem. Constraint-violation diagnostics at the optimum are at machine precision (maximum capacity violation 2.8×10^{-14} ; maximum demand violation 1.4×10^{-14}).

TABLE D.1. Transportation problems: dimensions and solutions

Problem	Cells	Providers	Variables	Time (s)	$T_{\min}$ (min)
Tract, home anchor (primary)	4,565	1,237	5,646,905	139.0	5.397
Block group, home anchor	9,215	1,237	11,398,955	508.6	5.470
Tract, corridor $\tau = 5$	4,565	1,237	5,646,905	175.5	1.552
Tract, corridor $\tau = 10$	4,565	1,237	5,646,905	173.5	1.342
Tract, corridor $\tau = 15$	4,565	1,237	5,646,905	172.9	1.282

D.3. Why strata are independent marginal problems. Stratum-level results (age, funding, Black Belt, region) are each solved as an *independent* transportation problem: the stratum’s own origin–age demand against the full fixed 1,237-provider capacity and eligibility. We first implemented the alternative of decomposing the joint statewide optimum’s flows by stratum, and rejected it after a quality-assurance discovery: the joint problem admits multiple exactly optimal allocations, so stratum shares of the joint flows depend on solver tie-breaking and are not identified. Bootstrap replication made this visible when several joint-decomposition stratum estimates fell outside their own resampling intervals; switching the estimand to independent marginal problems removed the problem. The statewide optimum itself is unaffected.

The independent-stratum estimand has a simple interpretation: each stratum’s $T_{\min}$ is the lower bound on that stratum’s mean travel if the entire fixed supply were available to it alone. Because strata do not compete for capacity in these problems, stratum minima need not aggregate to the joint statewide minimum, and stratum indices are conditional bounds, not an additive decomposition. An exploratory age-by-area-by-funding cross-tabulation using proportional joint-flow attribution exists in the replication materials for descriptive purposes only, with four small cells masked under the disclosure rule; it is not a primary estimand.

D.4. Scale sensitivity mechanics. The block-group problem re-solves the identical formulation with 3,356 origins (9,215 origin–age cells; 11,398,955 decision variables), holding observed exact-point costs fixed. Intrazonal assignments retain routed

centroid-to-provider times (never zero), so the tract-versus-block-group comparison isolates aggregation effects rather than an intrazonal accounting convention. Within-unit optimal shares are not comparable across scales because the units differ in size: the block-group optimum places 26.2% of children within their origin block group, and 45.3% within their home *tract*, against the tract optimum’s 49.1%. Age-stratum block-group re-solves reproduce the direction of every tract-level age contrast; because they predate the estimand correction of [Appendix D.3](#) they are read as directional sensitivity only.

A related grain asymmetry is quantifiable directly: observed costs use exact home coordinates while counterfactual costs use origin centroids. For the 37,991 dyads (99.8%) whose attended provider lies within the candidate prefilter, replacing each exact-home route with the tract-centroid route to the same attended provider shifts the observed mean from 13.07 to 13.28 minutes on the matched set, an offset of +0.21 minutes, or about 2.6% of the 8-minute excess. The exact-origin observed mean is therefore slightly *conservative* relative to a fully centroid-based design.

Appendix E. Bootstrap inference

E.1. Design. We quantify sampling uncertainty by a nonparametric bootstrap over dyads: $B = 1,000$ resamples of the 38,069 retained dyads, drawn with replacement under a fixed seed. Each replicate rebuilds origin–age demand from the resampled dyads, recomputes the observed mean T_{obs} , and re-solves the full capacity-constrained transportation problem, together with all 21 stratum problems (four age bands, two Black Belt areas, nine home regions, six funding groups), against fixed providers, licensed capacities, age-eligibility flags, and routed costs. The result is 22,000 replicate-by-stratum solutions. Intervals are percentile 95% intervals; no bias correction is applied. Estimated bootstrap bias for the statewide index is -0.12 percentage points, and every point estimate lies inside its own interval.

Held fixed across replicates are the supply side and the network: provider locations, capacities, eligibility, and all routed costs. The intervals therefore quantify demand-resampling variation conditional on measured supply, not uncertainty in capacity measurement, geocoding, routing, or household-level clustering (siblings enter as separate dyads).

E.2. Fast exact solver and benchmark. Re-solving the 5.6-million-variable primary problem 1,000 times with the reference solver (139 s per solve) is impractical. Replicates instead use a transportation-simplex implementation (`transport::transport`,

short-simplex method) on the capacity-balanced equivalent problem, in which a zero-cost dummy origin absorbs unused capacity. On the original sample the two solvers agree to $|\Delta T_{\min}| = 8.0 \times 10^{-15}$ minutes (0.45 s versus 139.0 s), i.e., to machine precision; the speedup makes the full-replication design feasible (total bootstrap wall time 487 s on three local workers).

E.3. Complete interval table. [Table E.1](#) reports point estimates, percentile intervals, and bootstrap standard errors for all 22 strata. [Figure E.1](#) displays the replicate distribution of the statewide index.

TABLE E.1. Excess-commuting index: point estimates and percentile bootstrap intervals ($B = 1,000$)

Dimension	Stratum	N	T_{ex} (%)	95% CI	SE (pp)
Overall	Statewide	38,069	59.80	59.18–60.14	0.25
Age	Infant (<1)	2,032	65.30	63.44–66.88	0.87
Age	Toddler (1–<3)	10,020	62.93	62.02–63.69	0.41
Age	Preschool (3–<6)	15,540	62.65	61.81–63.27	0.38
Age	School-age (6+)	10,477	63.68	62.76–64.52	0.46
Area	Black Belt (23 counties)	10,225	54.83	53.68–55.62	0.50
Area	Rest of Alabama	27,844	61.30	60.65–61.74	0.28
Region	Birmingham	8,282	70.41	69.31–71.25	0.50
Region	Mobile	9,025	60.10	59.17–60.84	0.42
Region	Montgomery	6,446	58.49	57.35–59.46	0.54
Region	Dothan	2,759	57.21	55.53–58.70	0.83
Region	Huntsville	4,316	52.88	51.67–54.03	0.61
Region	Ft. Payne	959	52.37	49.92–54.63	1.23
Region	Tuscaloosa	1,952	52.15	49.04–53.96	1.24
Region	Talladega	1,254	51.95	49.19–54.17	1.30
Region	Opelika	3,076	49.38	47.44–50.75	0.88
Funding	JOBS	1,221	66.76	64.70–68.48	0.98
Funding	Other	378	65.54	61.42–69.51	2.09
Funding	Early Head Start	515	64.74	61.89–67.44	1.42
Funding	At Risk	33,063	61.26	60.66–61.63	0.26
Funding	Protective Services	1,503	56.84	54.17–59.25	1.26
Funding	Foster Care	1,389	51.18	48.55–53.56	1.24

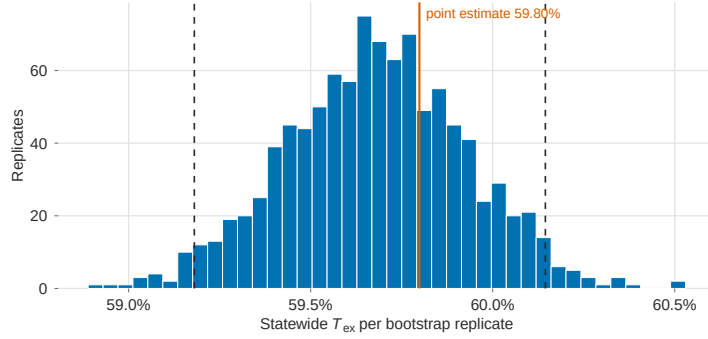


FIGURE E.1. Sampling distribution of the statewide excess-commuting index across 1,000 dyad-bootstrap replicates. The solid line marks the point estimate; dashed lines mark the 2.5th and 97.5th percentiles

E.4. Paired stratum contrasts. Because every stratum problem is re-solved within the same replicate, replicate-level paired differences are directly available and inherit the correlation structure of the resampled demand. [Table E.2](#) reports percentile intervals for the contrasts invoked in the main text. The infant index exceeds the toddler and preschool indices in nearly every replicate; the contrast with school-age children is positive in 94.6% of replicates but its interval crosses zero. The area and Birmingham–Mobile contrasts are positive in all 1,000 replicates.

TABLE E.2. Paired bootstrap contrasts of the excess index (percentage points; $B = 1,000$)

Contrast	Mean difference	95% CI	Share > 0
Infant – Toddler	2.4	0.4 to 4.3	0.991
Infant – Preschool	2.6	0.8 to 4.4	0.996
Infant – School-age	1.6	−0.4 to 3.4	0.946
Rest of Alabama – Black Belt	6.5	5.3 to 7.6	1.000
Birmingham – Mobile	10.3	9.0 to 11.5	1.000

E.5. History note. The first full bootstrap run used the joint-optimum flow decomposition for strata; its own diagnostics exposed the identification problem described in [Appendix D.3](#) (several stratum point estimates fell outside their resampling intervals). We discarded that run, corrected the estimand to independent marginal problems, and expanded the final 1,000-replicate run, which initially covered 13 strata, to all 22 strata so that region intervals could be reported. Only the corrected run enters any result in the main text or this supplement.

Appendix F. Dual-anchor work-corridor sensitivity

F.1. LODES processing and workplace centroids. Household workplaces are unobserved, so we build the work anchor as an area-level proxy from 2023 LODES origin–destination employment flows (LODES8, all-jobs series) (U.S. Census Bureau, 2026a). Five source files are combined (the Alabama main file plus the Florida, Georgia, Mississippi, and Tennessee auxiliary files, restricted to Alabama residents), each verified against a pinned SHA-256 fingerprint. The union covers 1,998,526 workers residing in the 1,373 demand tracts, of whom 5.30% work across a state border. Every demand tract has at least one flow.

For each home tract g , the work anchor W_g is the worker-weighted centroid of its workplace tracts, with workplace tract locations taken as interior points of the official 2020 polygons. A single dominant destination would argue for a different construction, but destinations are dispersed: the median tract sends only 8.5% of its workers to its largest single destination tract (maximum 38.7%), which supports the weighted-centroid summary but confirms that W_g is an ecological construct rather than any household’s workplace. The 2023 flow vintage also predates the subsidy extract; both facts argue for reading the corridor analysis as a bounding sensitivity.

F.2. Corridor cost construction. The excess-path detour of provider j relative to tract g ’s home–work axis is

$$q_{gj} = \frac{\|H_g - P_j\| + \|P_j - W_g\| - \|H_g - W_g\|}{v}, \quad v = 50 \text{ km/h}, \quad (\text{F.1})$$

floored at zero and expressed in minutes, where H_g is the home centroid and P_j the provider location. At tolerance τ , any provider with $q_{gj} \leq \tau$ receives generalized cost $\min(c_{gj}, q_{gj})$; all others keep the home-routed cost c_{gj} . The same generalized cost redefines the observed burden of each attended provider, so the corridor scenario changes both the numerator and the denominator of the normalized index. Because the modification can only lower costs, $T_{\min}^{\text{corridor}} \leq T_{\min}^{\text{home}}$ mechanically. The detour is Euclidean at a declared scenario speed; it does not reconstruct a routed trip chain.

F.3. Complete results. Table F.1 reports all quantities for the home-only anchor and the three tolerances, including the coverage diagnostics: the share of attended dyads whose provider lies on the corridor (and therefore has its observed cost redefined) and the share of all origin–provider pairs that gain a corridor cost. Even at $\tau = 15$, corridor pairs are under 3% of the candidate universe, yet they cover 79% of attended dyads. Families’ chosen providers are thus disproportionately near the

tract-level home-work axis, which is consistent with work-adjacent choice; absolute excess nevertheless falls only from 8.03 to 7.39 minutes.

TABLE F.1. Dual-anchor sensitivity: complete results. Corridor rows redefine both observed and minimum costs. “On corridor” is the share of attended dyads whose provider lies within the tolerance; “Pairs” is the share of all origin-provider pairs gaining a corridor cost

Scenario	T_{obs}	T_{min}	Excess (min)	T_{ex} (%)	On corridor (%)	Pairs (%)
Home only	13.43	5.40	8.03	59.80	—	—
Corridor $\tau = 5$	9.71	1.55	8.16	84.01	53.3	1.5
Corridor $\tau = 10$	8.98	1.34	7.63	85.05	70.1	2.3
Corridor $\tau = 15$	8.67	1.28	7.39	85.21	79.3	2.9

F.4. Why anchors are compared in absolute minutes. A natural prior is that admitting work-corridor options should lower both the minimum and the normalized index. The first holds by construction; the second need not, and empirically does not: the corridor index rises to about 85% because the re-optimized minimum falls proportionally faster than the redefined observed cost. The two normalized indices divide different numerators by different denominators and are therefore different estimands; ranking them against each other would misread a definitional artifact as a behavioral finding. Absolute excess (7.4 to 8.2 minutes across all specifications) is the quantity defined on a common scale, and it is the basis for the main text’s conclusion that a generous corridor allowance leaves the excess largely intact. This resolution follows the general caution that excess-commuting benchmarks are only comparable under a common cost definition (Kanaroglou et al., 2015), and it addresses the trip-chaining critique of home-anchored benchmarks (Hu & Li, 2021) at the level of aggregate sensitivity rather than household trip reconstruction.

Appendix G. Complete stratified results and the bypass diagnostic

G.1. Stratified excess commuting with cost decompositions. Table G.1 reports the complete stratified results (observed and minimum means, absolute excess, and the normalized index with intervals) for every stratum, including the nine home service regions summarized graphically in the main text (Fig. 5). All strata are independent marginal transportation problems (Appendix D); intervals are the percentile bootstrap intervals of Appendix E. Note that Foster Care’s low index reflects the highest minimum floor of any stratum ($T_{\text{min}} = 7.29$ minutes), since its placements face a thin eligible-supply geography, whereas JOBS’s high index reflects the lowest

floor (3.46 minutes) under short observed travel. Birmingham combines the longest observed mean among the nine regions (15.69 minutes) with the lowest regional minimum floor (4.64 minutes), which is why its index separates from every other region.

TABLE G.1. Complete stratified excess-commuting results (minutes; index in percent)

Dimension	Stratum	N	T_{obs}	T_{min}	Excess	T_{ex}	95% CI
Overall	Statewide	38,069	13.43	5.40	8.03	59.80	59.18–60.14
Age	Infant (<1)	2,032	14.79	5.13	9.66	65.30	63.44–66.88
Age	Toddler (1–<3)	10,020	14.07	5.22	8.86	62.93	62.02–63.69
Age	Preschool (3–<6)	15,540	13.74	5.13	8.61	62.65	61.81–63.27
Age	School-age (6+)	10,477	12.07	4.39	7.69	63.68	62.76–64.52
Area	Black Belt (23 counties)	10,225	11.60	5.24	6.36	54.83	53.68–55.62
Area	Rest of Alabama	27,844	14.10	5.45	8.64	61.30	60.65–61.74
Region	Birmingham	8,282	15.69	4.64	11.05	70.41	69.31–71.25
Region	Mobile	9,025	13.52	5.39	8.13	60.10	59.17–60.84
Region	Montgomery	6,446	11.31	4.69	6.61	58.49	57.35–59.46
Region	Dothan	2,759	12.31	5.27	7.04	57.21	55.53–58.70
Region	Huntsville	4,316	12.71	5.99	6.72	52.88	51.67–54.03
Region	Ft. Payne	959	14.56	6.93	7.62	52.37	49.92–54.63
Region	Tuscaloosa	1,952	14.65	7.01	7.64	52.15	49.04–53.96
Region	Talladega	1,254	15.08	7.25	7.84	51.95	49.19–54.17
Region	Opelika	3,076	11.68	5.91	5.77	49.38	47.44–50.75
Funding	JOBS	1,221	10.40	3.46	6.94	66.76	64.70–68.48
Funding	Other	378	15.87	5.47	10.40	65.54	61.42–69.51
Funding	Early Head Start	515	15.43	5.44	9.99	64.74	61.89–67.44
Funding	At Risk	33,063	13.35	5.17	8.18	61.26	60.66–61.63
Funding	Protective Services	1,503	14.87	6.42	8.45	56.84	54.17–59.25
Funding	Foster Care	1,389	14.93	7.29	7.64	51.18	48.55–53.56

An exploratory age-by-area-by-funding cross-tabulation (47 populated cells) exists in the replication materials with four cells below the disclosure threshold masked; it uses proportional joint-flow attribution and is descriptive only ([Appendix D.3](#)).

G.2. Bypass: definition and construction. Bypass is a trip-level revealed-use diagnostic defined as attended-provider routed time minus the routed time to the nearest participating, age-eligible provider from the same exact home. It requires no system-wide reallocation and therefore complements the transportation-problem results. The nearest-provider search shortlists, for each home’s block group, the 40 providers with the shortest centroid-routed times, then routes each exact home to its full shortlist; the attended provider is always included, so bypass is nonnegative by

construction. Shortlist adequacy checks: the chosen provider ranks beyond the 20th shortlist position in only 0.28% of dyads, and ten independent one-off re-routes match batch results exactly.

G.3. Bypass results. Statewide, mean bypass is 8.63 minutes (median 5.75; 90th percentile 18.51). Nearly all dyads bypass by a positive margin (84.9% exceed half a minute); 55.0% bypass by more than five minutes; and only 12.5% attend their nearest eligible provider or tie it. Where both quality tiers are observed, 13.1% of all dyads attend a provider with a strictly higher star tier than their nearest eligible alternative. [Figure G.1](#) shows the age pattern: infants bypass most in both minutes (9.59) and prevalence (60.2% over five minutes), echoing the stratum ordering of [Table G.1](#). Black Belt dyads bypass less (6.94 minutes; 50.5%) than the rest of the state (9.25; 56.6%), consistent with their shorter observed travel; Early Head Start dyads are the likeliest funding group to bypass by more than five minutes (63.5%; mean 9.72 minutes) and by far the likeliest to travel to a higher-tier provider (49.7%), consistent with program placements concentrating in rated facilities.

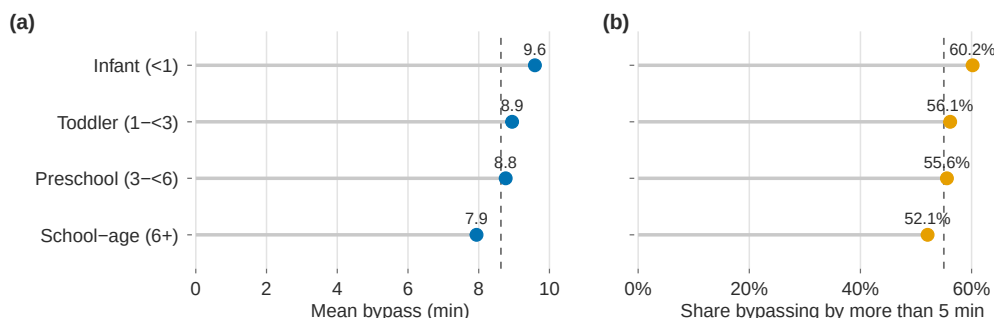


FIGURE G.1. Bypass burden by age group. (a) Mean bypass minutes beyond the nearest routed, participating, age-eligible provider. (b) Share of dyads bypassing by more than five minutes. Dashed lines mark statewide values (8.63 minutes; 55.0%)

G.4. A descriptive bypass model. [Table G.2](#) reports a logistic regression of meaningful bypass (more than five minutes) on age band, funding group, area, the attended provider's star tier, and the weekly copayment, on the 36,406 dyads with complete tier information. The model is descriptive; coefficients bundle geography, eligibility, and selection. Three patterns are robust: infant dyads have 42% higher odds of meaningful bypass than school-age dyads; residence outside the Black Belt raises the odds by 30%; and placement-driven funding categories (Foster Care, Protective Services)

show *lower* bypass odds than At Risk, consistent with administratively arranged rather than searched placements. Tier main effects are small and mostly indistinguishable from Star 1, reinforcing that measured tier differences do not account for most bypass.

TABLE G.2. Logistic regression of bypass exceeding five minutes ($n = 36,406$ dyads with observed tiers). Reference categories: School-age (6+), At Risk, Black Belt, Star 1

Term	Odds ratio	Log-odds (SE)	p
Intercept	0.916	-0.088 (0.028)	0.001
Infant (<1)	1.418	0.350 (0.051)	<0.001
Toddler (1-<3)	1.152	0.141 (0.029)	<0.001
Preschool (3-<6)	1.126	0.118 (0.027)	<0.001
Early Head Start	1.340	0.292 (0.097)	0.003
Foster Care	0.790	-0.236 (0.057)	<0.001
JOBS	0.961	-0.040 (0.060)	0.507
Other	0.926	-0.077 (0.108)	0.474
Protective Services	0.772	-0.259 (0.056)	<0.001
Rest of Alabama	1.302	0.264 (0.024)	<0.001
Star 2	0.930	-0.072 (0.038)	0.058
Star 3	0.916	-0.088 (0.053)	0.095
Star 4	0.894	-0.112 (0.048)	0.019
Star 5	1.014	0.014 (0.092)	0.879
Weekly copayment (per \$1)	1.003	0.003 (0.001)	<0.001

Appendix H. Opportunity-adjusted travel-time decay

H.1. Exposure construction and likelihood. Observed travel-time histograms confound willingness to travel with the geography of available supply, so we model utilization *per opportunity*. For stratum s and five-minute bin b , the exposure e_{sb} sums, over demand tracts, the tract-age demand multiplied by the count of participating, age-eligible providers whose tract-centroid routed time falls in bin b ; the outcome n_{sb} counts observed dyads whose exact-home routed time falls in the same bin. The pooled fit compares 37,900 observed trips against 8,969,991 demand-weighted opportunity exposures. Bins mix an exact-point numerator with a centroid-based denominator; the five-minute width absorbs the resulting within-bin grain mismatch.

Counts are modeled by Poisson regression with $\log e_{sb}$ as offset,

$$n_{sb} \sim \text{Poisson}(e_{sb} \mu_s(t_b)), \quad (\text{H.1})$$

with three mean functions of bin midpoint t_b : exponential $\mu(t) = Ae^{-\lambda t}$, power $\mu(t) = A(t+1)^{-\alpha}$, and Gaussian $\mu(t) = Ae^{-\beta t^2}$. Fitting uses $0 \leq t < 90$ minutes, within

which candidate completeness is guaranteed by the routing prefilter (Appendix C.3); the 169 observed trips at 90 minutes or beyond (0.44%) are excluded from fitting only. Standard errors are scaled by Pearson overdispersion (pooled dispersion 108.8), so the fitted curves are read as parsimonious descriptive summaries with conservative intervals rather than as a fully specified count process.

H.2. Functional-form comparison. The exponential form is preferred by Poisson AIC and by log-rate RMSE in the pooled sample and within every age stratum (15 of 15 fits converged). Table H.1 reports the pooled comparison; the power form’s nominal half-life below one minute and the Gaussian’s unstable scaled interval (its lower decay bound crosses zero) are symptoms of misfit, not competing calibrations.

TABLE H.1. Pooled functional-form comparison (fit window $0 \leq t < 90$ minutes)

Form	Decay parameter	Half-life (min)	AIC	Δ AIC	Log-rate RMSE
Exponential	$\lambda = 0.1062$	6.53	1,719.0	0	0.652
Power	$\alpha = 1.606$	0.54	5,614.2	3,895.2	1.586
Gaussian	$\beta = 0.00219$	17.78	10,932.7	9,213.7	3.951

H.3. Age-specific parameters. Table H.2 reports the exponential fits by age band with overdispersion-scaled 95% intervals. Half-lives decline monotonically with age: infants’ relative use of farther opportunities decays most slowly, school-age children’s most quickly. These are opportunity-adjusted revealed-use associations under the current supply configuration, not causal effects of age.

TABLE H.2. Age-specific exponential decay parameters (overdispersion-scaled 95% CIs)

Stratum	λ (per min)	95% CI for λ	Half-life (min)	95% CI for half-life
Pooled	0.1062	0.0943–0.1181	6.53	5.87–7.35
Infant (<1)	0.0899	0.0800–0.0998	7.71	6.95–8.66
Toddler (1–<3)	0.0992	0.0877–0.1108	6.99	6.26–7.91
Preschool (3–<6)	0.1040	0.0933–0.1148	6.66	6.04–7.43
School-age (6+)	0.1237	0.1086–0.1388	5.60	4.99–6.38

H.4. Comparison with imposed conventions. Table H.3 evaluates the fitted pooled curve against three conventions used in child care accessibility work: the Luo–Qi-style three-zone step (1.00/0.68/0.22 with a 15-minute boundary) (Luo & Qi, 2009), the $\exp(-t^4/1000)$ decay of Davis et al. (2019) capped at 20 minutes (verified

against the source’s technical appendix), and a hard 15-minute cutoff. Over 0–60 minutes, the step function’s total weight area (9.63) nearly matches the empirical curve’s (9.40), but the step reallocates that weight toward the first ten minutes and places 0.3% of its area beyond 15 minutes where the empirical curve places 20.2%. The Davis form is far steeper than revealed use throughout (total area 5.10); the hard cutoff overweights everything inside 15 minutes (area 15.13) and zeroes the 29.7% of observed trips beyond it. Exposure-weighted mean absolute deviations from the empirical curve are 0.046 (step), 0.074 (Davis), and 0.121 (hard cutoff).

TABLE H.3. Relative accessibility weight at selected travel times: fitted curve versus imposed conventions

t (min)	Empirical $e^{-0.1062t}$	Three-zone step	Davis $\beta=4$	Hard 15-min
0	1.000	1.00	1.000	1
5	0.588	1.00	0.535	1
10	0.346	0.68	4.5×10^{-5}	1
15	0.203	0.22	1.0×10^{-22}	1
20	0.120	0	3.3×10^{-70}	0
30	0.041	0	0	0
45	0.008	0	0	0
60	0.002	0	0	0

H.5. Exported parameters. The replication materials export a machine-readable parameter file containing the pooled and age-specific exponential parameters with their intervals, the derived half-lives and 5%-weight points, and exact specifications of the three comparison conventions with source attributions. The intended use is as drop-in accessibility weights or as a sensitivity band around any imposed catchment. The export is a descriptive calibration of revealed use under the current opportunity structure, not a normative access standard. Because it is estimated on enrolled participants, it plausibly understates the distance sensitivity of the full eligible population, and it should be re-estimated where the supply geography differs materially.

Appendix I. Privacy, disclosure control, and reproducibility

I.1. Governance and the restricted–public boundary. Child-level records are governed by a restricted data sharing agreement with the Alabama Department of Human Resources (identifier on the title page) and applicable state confidentiality law. All row-level data (dyads, exact coordinates, and routes) reside in access-controlled local storage, are excluded from version control, and never leave the analysis machine; the routing engine runs on localhost precisely so that no coordinate is transmitted

externally (Appendix C.1). Direct identifiers are dropped at ingest; the analytic dyad identifier is a row sequence with no linkage value outside the restricted environment.

Public outputs are aggregates only, under a minimum-cell rule: any released cell with fewer than 10 observations is suppressed or masked. Applied instances in this project include two county rows in the observed-travel distribution table, four cells of the exploratory cross-tabulation (Appendix G.1), and histogram bins below the threshold (main text, Fig. 2). No figure or map displays child locations at any scale; maps show administrative aggregates and public licensed-business points only, without provider identifiers, names, or addresses. An automated string scan of every public artifact for identifier fields returns zero hits.

I.2. Computational environment. We ran all analyses in R 4.6.0 on Apple-silicon macOS. Key components: OSRM backend 26.7.3 (multi-level Dijkstra, car profile) with the `osrm` R client 5.0.0 for health checks; `sf` 1.1-1 and `tigris` 2.2.1 for spatial operations against 2020 cartographic geography; `lpSolve` (transportation simplex) for reference optimizations and `transport` (short-simplex) for bootstrap re-solves; `data.table` and `dplyr` 1.2.1 for data assembly; `ggplot2` for figures. The heaviest single jobs are the block-group optimization (509 s), the three corridor optimizations (173–176 s each), the primary tract optimization (139 s), and the 1,000-replicate bootstrap (487 s on three workers). LODS downloads are pinned by SHA-256; the road-network extract is pinned by MD5 and SHA-256 (Appendix C.1).

I.3. Reproducibility and claim verification. The pipeline is a fixed sequence of scripts from ingest to manuscript-support outputs, with three verification layers. First, a registry records every reported value with its generating script and timestamp; figures and tables are rebuilt only from analytic objects and are machine-reconciled against the registry (maximum absolute discrepancy across checked values: 3.6×10^{-14}). Second, a claim inventory maps each substantive claim to its evidence file and caveat, and an automated checker verifies every numeric claim in the manuscript against the registry (26 of 26 pass). Third, solver-level checks (integrality, constraint-violation magnitudes, benchmark agreement between independent solver implementations; Appendices D and E) check the optimization layer itself. The public replication package—the analysis code, source manifests, the aggregate disclosure-reviewed outputs (excess-commuting and bypass statistics, the stratum tables, and the fitted travel-time decay parameters), and the publication figures and tables—is available at <https://github.com/joonho112/childcare-excess-commuting-replication>.

The child-level administrative records are governed data provided by the Alabama Department of Human Resources under a restricted data sharing agreement and are not redistributed in the package; the aggregate outputs it contains are sufficient to reproduce every reported result.